SHRIHARIHARAN S
AI/ML Engineer | Generative AI |Machine Learning | Data Science
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PROFESSIONAL SUMMARY

AI/ML and Generative AI Engineer with 4+ years of professional IT experience, including enterprise payment implementation and test engineering at Wipro, an M. Tech in Computer Science (AI & Data Science) from VIT Chennai (CGPA 9.28/10), and hands-on experience building AI applications. I work primarily with Python and have built machine learning and deep learning solutions, QLoRA fine-tuned LLMs, hybrid RAG systems using FAISS and BM25 with reranking, and multi-agent applications using LangChain, LangGraph, MCP, and FastAPI. My recent work includes VoyageAI, a live multi-agent travel planner deployed on Google Cloud, an autonomous data analyst for CSV/Excel data, an Enterprise RAG platform, and an AI automation testing agent. At Wipro, I owned 30+ Mastercard payment-modernization projects across requirements, implementation, testing, and production rollout. During my Intel internship, I worked on Python-based automation for post-silicon validation, cache coherency, seamless runtime patching of processor microcode components such as uCode/pCode, structured log analysis, and failure investigation, while also building RAG-based and agentic AI applications. I enjoy turning AI concepts into practical systems with APIs, evaluation, guardrails, testing, and deployment.

TECHNICAL SKILLS

- **Generative AI & Agentic Systems:** LLMs, RAG, Hybrid Retrieval (FAISS + BM25), Cross-Encoder Reranking, LangChain, LangGraph, deepagents, MCP, Tool Calling, Multi-Agent Systems, Human-in-the-Loop, Guardrails, Self-Healing, LLM-as-a-Judge, LLM Evaluation, Prompt Engineering, LoRA, QLoRA, PEFT, LangSmith
- **Machine Learning & Data Science:** Regression, Classification, Decision Trees, Random Forest, SVM, KNN, XGBoost, LightGBM, CatBoost, AdaBoost, Gradient Boosting, K-Means, DBSCAN, PCA, Feature Engineering, Hyperparameter Tuning, Cross-Validation, SHAP, SMOTE, Data Analysis & Visualization, ML Pipelines
- **Deep Learning & NLP:** ANN, CNN, RNN, LSTM, GRU, Transfer Learning, Computer Vision, Time-Series Forecasting, GANs, Transformers, Self/Multi-Head Attention, Tokenization, Embeddings, TF-IDF, Word2Vec, NER, Sentiment Analysis
- **Languages & Frameworks:** Python, SQL, C, PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, NLTK, Hugging Face Transformers, Sentence-Transformers, Pydantic, Streamlit
- **Deployment, Cloud & APIs:** FastAPI, REST APIs, Uvicorn, AsyncIO, Docker, Google Cloud Run, Firebase (Auth, Firestore, Hosting), Apache Kafka, MySQL, SQLite, Git, Postman, Pytest, Model Deployment · OpenAI, Google Gemini, NVIDIA AI APIs, Groq, Ollama

PROFESSIONAL EXPERIENCE

Test Engineer, Wipro Limited

Chennai, Tamil Nadu | ***Sep 27, 2021 - Aug 10, 2026*** | *Sabbatical Jun 2024 - Aug 2026 for full-time M. Tech; relieved after graduation*

- Delivered 30+ Mastercard payment modernization projects across North America, APAC, Latin America, and EMEA, covering implementation, testing, and production rollout.
- Executed BIN 8 transition testing by validating customer-provided retain and return ranges and performing mainframe SQL checks against approved migration requirements.
- Validated 180-day transaction activity, including SMS and DMS transactions, and assessed mainframe dependencies to determine whether customer approval or further investigation was required.
- Supported Token Replenishment by validating ICA and card details, reviewing eligible Token Pool ranges, and verifying range assignments against applicable business rules before MDES System Manager configuration.
- Explored AI-assisted testing to suggest additional test scenarios, compare candidate ranges against approved requirements, and explain validation exceptions, with final verification performed using SQL results and established business rules.

Graduate Technical Intern (Internship), Intel Corporation

Bangalore, Karnataka | ***Jul 2, 2025 - May 27, 2026***

- Built RAG-based and Agentic AI applications using Lang Chain, multi-agent workflows, and retrieval-augmented generation during Intel's internal AI learning programs and Hackathon.
- Performed post-silicon validation of Intel Xeon (Forest Series) server processors, focusing on cache coherency and system-level validation.
- Validated live-patching (Seamless Patching) mechanisms for Rapid Series processors across multiple runtime scenarios while maintaining platform stability.
- Developed Python automation scripts to accelerate OOBMSM validation, reducing manual testing effort through automated workflows and for Validation.
- Conducted structured log analysis, debugging, and failure investigation to identify and isolate hardware and software validation issues. Worked with enterprise validation processes, improving reliability and efficiency of large-scale server testing.

AI/ML, GENERATIVE AI & DATA SCIENCE PROJECTS

VoyageAI: AI Tour Planner, Live Full-Stack Multi-Agent Travel Assistant

FastAPI, LangChain, LangGraph, deepagents, Gemini, React, Firebase Auth, Firestore, Docker, Google Cloud Run, LangSmith

Live: voyageai-702fe.web.app | **Code:** github.com/Shrihariharan1999/Agentic_AI/tree/main/ai-tour-planner

- Built a multi-agent travel planner that turns natural-language requests into date-aware, day-by-day itineraries; a planner agent orchestrates a weather sub-agent and 8 live-data tools (Google Places/Routes, flights, hotels, train availability, weather, web search).
- Grounded and controlled the agents: prices, URLs, and schedules come only from tool results and the agent stops to ask the user instead of guessing when constraints cannot be met.
- Secured the FastAPI backend with Firebase ID-token verification and per-user ownership checks; chats persist in Firestore.
- Deployed React on Firebase Hosting (Google Sign-In) and a Dockerized FastAPI backend on Cloud Run, traced with LangSmith.

Agentic AI Platform: Autonomous Data Analyst, Web Testing & MCP Tool-Using Agents

LangGraph-style agents, FastAPI, Streamlit, SQLite, FastMCP, Playwright, Gemini, LangSmith · github.com/Shrihariharan1999/Agentic_AI

- **Autonomous AI Analyst:** Built a multi-agent application with Coordinator, Analyst, and Validator agents that converts natural-language questions on CSV/Excel datasets into statistical analysis, visualizations, and web-researched answers, with SQLite for session and dataset-version tracking. Developed an LLM-as-a-Judge validation workflow that reviews generated reports, identifies issues, and automatically retries the draft before producing the final Markdown/HTML report, with file access sandboxed through a custom MCP server.
- **AI Automation Testing:** Built a Playwright-based web-testing agent supporting human-in-the-loop approval, self-healing retries, and LangSmith tracing for browser-based test execution.
- **AI Code Interpreter:** Developed a separate Python-based code-interpreter application within the same repository for executing AI-generated analytical/code workflows in response to user requests.

Enterprise RAG System

Python, LangChain, FAISS, BM25, Sentence-Transformers, Gemini · github.com/Shrihariharan1999/Enterprise_RAG

- Built a modular RAG pipeline (ingestion, chunking, vector store, retrieval) supporting 8 document formats, combining FAISS semantic search with BM25 keyword search and cross-encoder reranking.
- Added spell-correction(optional), duplicate-file detection, page-level citations, and persistent chat history for traceable answers.

LLM Fine-Tuning with QLoRA

Python, PyTorch, Hugging Face Transformers, PEFT, QLoRA · github.com/Shrihariharan1999/LLM_FineTuning_QLORA_2026

- Fine-tuned open-source LLMs with QLoRA (4-bit quantization plus LoRA adapters), training only a small slice of parameters so the job ran on a single consumer GPU instead of a cluster.
- Benchmarked fine-tuned outputs against the base model to quantify the actual gain from adaptation.

Machine Learning, Deep Learning & NLP Portfolio

Python, Scikit-learn, TensorFlow, Keras, PyTorch, NLTK, LangChain

GitHub: Machine_Learning_MTECH | Deep_Learning_2026 | Machine_Learning_2026

- 87-notebook, 11-module portfolio spanning supervised/unsupervised ML, ensembles, and CNNs/LSTMs/GANs/Transformers implemented from scratch (attention and positional encoding by hand).
- Includes a generative-AI module (prompt engineering, LoRA, RAG, LLM evaluation, GenAI ethics) and applied case studies: Titanic survival, cirrhosis-outcome prediction, and English-to-French translation.

MindMate: AI Mental Health Support Platform

Next.js 14, FastAPI, Gemini 2.5 Flash, Llama 3.3 (Groq), DeepFace/MTCNN · github.com/Shrihariharan1999/MindMate

- The frontend runs on Next.js with a FastAPI backend, JWT auth, and real-time WebSocket chat. Llama 3.3, accessed through Groq, acts as an LLM judge that scores each conversation for safety, honesty, and empathy, with a rule-based fallback if the model call fails. Gemini chat, DeepFace/MTCNN emotion detection, VADER sentiment, and Edge-TTS voice feed a Recharts mood dashboard.

HACKATHON

DATASET'24 Hackathon: Telecom Energy Optimization (Sponsored by Prodapt and partnered with Nokia), VIT Chennai

Python, Apache Kafka, LLMs, React | **Certificate:** Drive Link

- Built a real-time Kafka pipeline and LLM-driven model to predict network load and auto-switch cell towers to power-saving mode.
- Delivered in a 4-person team over a 36-hour hackathon and presented to industry judges.

EDUCATION & ACHIEVEMENTS

M. Tech, Computer Science & Engineering (AI & Data Science) | *2024 - 2026*

Vellore Institute of Technology, Chennai · CGPA: 9.28/10 · Meritorious Award for academic performance, Semester 1 and Semester 2

B.E., Electronics and Instrumentation Engineering | *2017 - 2021*

Panimalar Engineering College · CGPA: 8.66/10